

Which Rung?

84-355 Democracy's Data | Levels of aggregation in election data

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August 11, 2026

Everything here runs as given. Your job is to read what comes out and say what it means.

Read `levels-of-aggregation-brief.Rmd` first. This lab runs the same numbered steps with the code exposed. The brief tells you what the question is and why it is hard; this shows you where each number came from — and in Step 6 it *runs* the method that fails, rather than reporting that it failed.

Which unit is your claim about?

“Turnout was up in the suburbs.”

Every sentence like that names a level of aggregation without saying so. The suburbs are made of precincts, which are made of ballots, and they sit inside counties, districts, states and a country. **Six of those seven levels are available to you. One almost never is.**

This lab walks a single election up the ladder and shows what each step costs.

Step 1 — The ladder

```
ld <- read.csv("data/ladder.csv", stringsAsFactors = FALSE)
ld[, c("rung", "level", "units_nationally", "units_in_georgia", "a_row_is")]
```

#>	rung	level	units_nationally	units_in_georgia
#> 1	0	ballot	158211780	NA
#> 2	1	precinct	177708	2701
#> 3	2	election jurisdiction	6461	NA
#> 4	3	county	3144	159
#> 5	4	congressional district	434	14
#> 6	5	state	51	1
#> 7	6	nation	1	NA
#>		a_row_is		
#> 1		one voter's ballot		
#> 2		one polling place's returns		

```
#> 3 one office that runs elections
#> 4         one county's returns
#> 5     one seat in the U.S. House
#> 6         one state's returns
#> 7         the whole electorate
```

158,211,780 ballots. 177,708 precincts. 6,461 election jurisdictions. 3,144 counties. 434 congressional districts. 51 states. One nation.

Read the `published_for` column too — it is the column that decides what you can actually do.

```
ld[, c("level", "published_for")]
```

```
#>           level                                published_for
#> 1           ballot                                a handful of jurisdictions
#> 2           precinct                            every state, in different formats
#> 3 election jurisdiction                            every state, once every two years
#> 4           county                                every state except Alaska
#> 5 congressional district reconstructed from blocks by third parties
#> 6           state                                every state
#> 7           nation                                the country
```

Step 2 — The bottom rung, which barely exists

Alaska runs ranked-choice elections. To run them it must publish a **cast vote record**: one row per ballot, carrying every ranking on it.

```
cf <- read.csv("data/cvr_facts.csv", stringsAsFactors = FALSE)
cf
```

```
#>           quantity    value
#> 1           ballots in the file 340778.0
#> 2           distinct ranking sequences    926.0
#> 3           ballots with no ranking at all 10150.0
#> 4           ballots using exactly one ranking 215501.0
#> 5 bullet-voting rate among ballots that ranked anyone (%)    65.2
#> 6           ballots that named a second choice 115127.0
#> 7           ballots that ranked the same candidate more than once    5286.0
#> 8           ranked Begich first and Peltola second    6663.0
#> 9           ranked Begich first and Howe second    24386.0
#> 10          ranked Peltola first and Begich second    10771.0
#> 11          ranked Peltola first and Hafner second    39684.0
```

340,778 ballots. 926 distinct ranking sequences. 65.2% of voters who ranked anyone ranked exactly one candidate and stopped, which is why so few distinct sequences cover so many ballots.

Now the numbers that matter for this lab:

```
c2 <- read.csv("data/cvr_second.csv", stringsAsFactors = FALSE)
c2 <- c2[c2$second != "(no second choice)", ]
head(c2[order(-c2$ballots), ], 8)
```

```

#>           first           second ballots
#> 30 Peltola, Mary S.   Hafner, Eric   39684
#> 2   Begich, Nick Howe, John Wayne  24386
#> 31 Peltola, Mary S. Howe, John Wayne  13890
#> 32 Peltola, Mary S.   Begich, Nick   10771
#> 3   Begich, Nick Peltola, Mary S.    6663
#> 16 Howe, John Wayne   Begich, Nick   4017
#> 4   Begich, Nick     Hafner, Eric    3427
#> 5   Begich, Nick     Begich, Nick    2455

```

6,663 Alaskans ranked Begich first and Peltola second — a Republican first and a Democrat second, on one ballot. 10,771 did the reverse.

No total, at any level above this one, contains those numbers. They are properties of a *pair* of marks on one piece of paper, and adding the paper up destroys the pair.

Step 3 — One rung up

```

BALLOTS <- ld$units_nationally[ld$level == "ballot"]
AK      <- cf$value[cf$quantity == "ballots in the file"]
c(alaska_cvr_ballots = AK, all_2024_ballots = BALLOTS,
  pct_of_national = round(100 * AK / BALLOTS, 3))

```

```

#> alaska_cvr_ballots  all_2024_ballots  pct_of_national
#>           340778.000      158211780.000           0.215

```

The whole Alaska file is 0.215% of the ballots cast in 2024. There is no national cast-vote-record file. If you want to say something about the country, you must leave the bottom rung.

Steps 4 and 5 — Climb, and watch what holds

One election, the 2024 presidential vote in Georgia, read at four levels:

```

lst <- read.csv("data/level_stats.csv", stringsAsFactors = FALSE)
lst

```

```

#>           level units mean_dem_share sd_dem_share min_dem_share
#> 1 precinct    2684      48.90      24.68         3.4
#> 2 county      159      48.90      20.64         8.0
#> 3 congressional district    14      49.78      18.45        31.2
#> 4 state         1      48.89       0.00        48.9
#> max_dem_share variance_retained_pct
#> 1      100.0           100.0
#> 2      84.8           69.9
#> 3      86.4           55.9
#> 4      48.9           0.0

```

The vote-weighted mean does not move — 48.90% across precincts, 48.90% across counties, 48.89% for the state. The district row reads 49.78 because the district file carries percentages and no vote counts, so that one mean is unweighted. The spread collapses: the standard deviation of Democratic share falls from 24.68 points across 2,684 precincts to 20.64 across 159 counties. `variance_retained_pct` says that 69.9% of the precinct-level variation survives the trip to counties, so 30.1% of it does not.

Does the arithmetic hold?

```
cs <- read.csv("data/consistency.csv", stringsAsFactors = FALSE)
cs[, c("comparison", "from", "against", "difference")]
```

```
#>
#>      comparison      from
#> 1 Georgia Democratic votes 2,701 precincts summed
#> 2 Georgia Democratic votes 159 counties, state source
#> 3 Georgia Republican votes 159 counties, state source
#> 4 National Democratic votes 3160 counties summed
#> 5 National Republican votes 3160 counties summed
#>
#>      against difference
#> 1      159 counties, same source      0
#> 2 159 counties, national compilation    -11
#> 3 159 counties, national compilation    -7
#> 4      certified national total  -10,874
#> 5      certified national total  -7,781
```

Summing all 2,701 precincts reproduces the county file **exactly**. Against a different compiler the same Georgia counties are 11 votes off, and the whole country's counties land 10,874 votes short of the certified national total — 0.0145% low.

Part of the reason is that a rung is not a rung everywhere:

```
nc <- read.csv("data/not_counties.csv", stringsAsFactors = FALSE)
nc
```

```
#>
#>      state rows_that_are_not_counties
#> 1      Alaska                      37
#> 2 District of Columbia              7
#>
#>      what_the_row_actually_is
#> 1 State House District 1 ... State House District 9
#> 2      Ward 2 ... Ward 8
```

Alaska has no counties. The national county file carries 37 state house districts in their place, and DC contributes 7 wards.

And one rung cannot be reached by adding at all — precinct returns record no congressional district. But it is the only rung that can answer this:

```
sv <- read.csv("data/seat_vote.csv", stringsAsFactors = FALSE)
sv
```

```
#>
#>      measure value
#> 1 Georgia two-party Democratic share, from the precincts 48.90
#> 2      Georgia: unweighted mean of the 14 district shares 49.78
#> 3      Georgia: median district share 40.73
```

```
#> 4      National two-party Democratic share, certified 49.25
#> 5      National: unweighted mean of the 434 district shares 49.94
#> 6      National: median district share 48.44
#> 7      Trump's share of the national popular vote (%) 49.80
#> 8      Trump's share of the electoral college (%) 57.99
```

Georgia's median congressional district is **40.7% Democratic** while the state is **48.9%**. Trump won 49.8% of the national popular vote and 58.0% of the electoral college. Neither fact is visible from any other level.

Predict before you run Step 6. Georgia publishes its precinct returns broken out by vote method. Step 6 deletes that breakdown, keeps only each precinct's two margins, and runs the standard method used in voting-rights litigation to recover what share of mail voters chose Biden. **Write down your guess — and write down the largest value the answer could possibly take.**

Step 6 — Run the method that fails

First the truth, which Georgia published:

```
mt <- read.csv("data/method_truth.csv", stringsAsFactors = FALSE)
mt

#>      candidate by_mail not_by_mail  total
#> 1 Donald J. Trump  451157    2010680 2461837
#> 2   Jo Jorgensen   16057      46081   62138
#> 3 Joseph R. Biden  849729    1624778 2474507
#>  pct_of_this_candidates_votes_by_mail
#> 1
#> 2
#> 3

TRUE_MAIL <- 100 * mt$by_mail[mt$candidate == "Joseph R. Biden"] / sum(mt$by_mail)
TRUE_NOT  <- 100 * mt$not_by_mail[mt$candidate == "Joseph R. Biden"] /
              sum(mt$not_by_mail)
round(c(dem_share_of_mail_votes = TRUE_MAIL,
        dem_share_of_everything_else = TRUE_NOT), 2)

#>      dem_share_of_mail_votes dem_share_of_everything_else
#>      64.52                44.13
```

Now throw the crosstab away. Keep only what every other state publishes — one row per precinct, two margins — and fit Goodman's ecological regression:

```
ep <- read.csv("data/eco_precinct.csv", stringsAsFactors = FALSE)
head(ep[, c("unit", "votes", "mail_pct", "dem_pct")], 4)

#>      unit votes mail_pct  dem_pct
#> 1 Appling|1B   993 17.11984 11.681772
#> 2 Appling|1C   640 11.71875  6.875000
#> 3 Appling|2  1448 25.69061 62.707182
#> 4 Appling|3A1   611 12.27496  3.273322
```

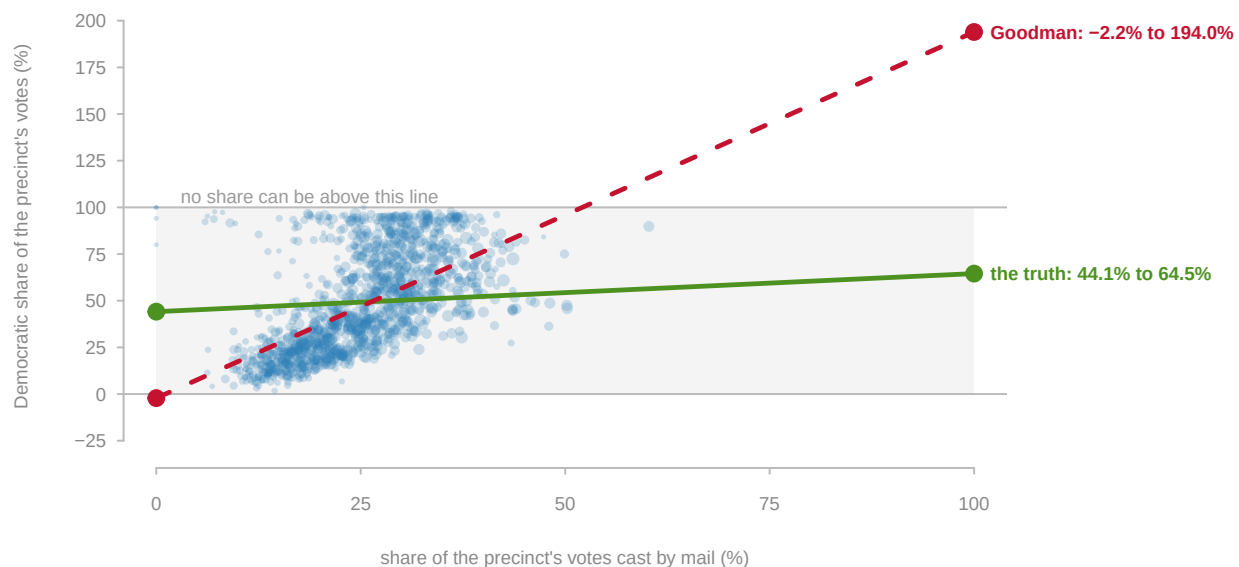
```
fit <- lm(dem_pct ~ mail_pct, data = ep, weights = ep$votes)
b <- coef(fit)
round(c(estimate_at_0_pct_mail = b[[1]],
        estimate_at_100_pct_mail = b[[1]] + 100 * b[[2]]), 1)
```

```
#> estimate_at_0_pct_mail estimate_at_100_pct_mail
#> -2.2 194.0
```

194.0%.

The regression, run on real precinct returns with a respectable method, says that 194.0% of Georgia's mail voters chose Biden. The truth is 64.5%. Nothing in the aggregate data announces the failure — the fit is fine and the line runs through the middle of the cloud.

2,653 Georgia precincts, 2020



Both lines pass through the same cloud. **They differ only in slope, and the slope is the whole answer.** Precincts that mailed heavily were Democratic precincts, and Democratic precincts were more Democratic among mail voters *and* election-day voters alike. The regression charges that entire between-precinct difference to the method of voting.

Step 7 — Do it again one rung higher

```
ec <- read.csv("data/eco_county.csv", stringsAsFactors = FALSE)
b2 <- coef(lm(dem_pct ~ mail_pct, data = ec, weights = ec$votes))
round(c(precinct_level = b[[1]] + 100 * b[[2]],
        county_level = b2[[1]] + 100 * b2[[2]],
        the_truth = TRUE_MAIL), 1)
```

```
#> precinct_level county_level the_truth
#> 194.0 233.4 64.5
```

Coarser units, worse answer: 233.4%. The full comparison, with Duncan and Davis's assumption-free bounds alongside:

```
ee <- read.csv("data/eco_estimates.csv", stringsAsFactors = FALSE)
ee

#>      level units correlation est_by_mail est_not_by_mail bound_lower
#> 1 precinct 2653      0.599      194.0        -2.2         15.2
#> 2 county  159      0.760      233.4       -16.3          6.6
#>      bound_upper truth_by_mail truth_not_by_mail error_by_mail
#> 1      98.4         64.5         44.1         129.5
#> 2      99.4         64.5         44.1         168.9
```

The bounds are the honest thing the aggregate *can* support. They are also nearly useless — 15.2% to 98.4% — and they get wider, not narrower, one rung up: 6.6% to 99.4%.

Is the county correlation a fact about voters or about county lines? Regroup the same precincts into the same number of *random* groups, 500 times:

```
mp <- read.csv("data/maup_null.csv", stringsAsFactors = FALSE)
mp

#>                                grouping units correlation
#> 1                2,653 precincts, ungrouped 2653      0.599
#> 2                159 real counties      159      0.760
#> 3 159 random groups of precincts (mean of 500 draws) 159      0.605
#>                                note
#> 1                the finest unit with published returns
#> 2                the same ballots, grouped by county line
#> 3 range 0.439 to 0.739; 0 of 500 draws reach the county figure
```

Random groups leave the correlation where it was (0.605 against 0.599 ungrouped). Real counties push it to 0.760, which none of the 500 draws reached. The lift comes from these particular lines — Openshaw's modifiable areal unit problem, which *areal-units* takes on by itself.

Step 8 — The same failure, where it counts

```
rp <- read.csv("data/rpv_check.csv", stringsAsFactors = FALSE)
rp

#>                                quantity value
#> 1                Black voters in the 17 precincts 4631.0
#> 2                Black voters who took a Democratic ballot 4293.0
#> 3                the truth: Democratic share of Black voters (%) 92.7
#> 4 Goodman's ecological regression on the same precincts (%) 102.1
#> 5                overshoot, in percentage points 9.4
```

rpv runs this on the question ecological inference is actually used for: **Goodman says 102.1% of Black voters took a Democratic ballot; the voter file says 92.7%.** Same method, same impossibility, smaller margin. *vote-dilution* takes the same county on into the first *Gingles* precondition.

Steps 9 and 10 — What the aggregate is for, and how to choose

```
dc <- read.csv("data/decision.csv", stringsAsFactors = FALSE)
dc[, c("question", "right_rung")]
```

```
#>                                     question
#> 1 Did Begich voters who ranked a second choice prefer Peltola or Howe?
#> 2           What share of a candidate's votes arrived by mail?
#> 3           Is voting in this county racially polarized?
#> 4           Which neighbourhoods swung between two elections?
#> 5           Which doors should a campaign knock on Saturday?
#> 6           How many mail ballots were rejected, and for what reason?
#> 7           Did counties with less education move right?
#> 8           Does this map convert votes into seats fairly?
#> 9           Who won the presidency?
#> 10          Is the country's partisan balance shifting across decades?
#>                                     right_rung
#> 1 ballot
#> 2 ballot, or precinct-by-method returns
#> 3 precinct
#> 4 precinct
#> 5 precinct
#> 6 election jurisdiction
#> 7 county
#> 8 congressional district
#> 9 state
#> 10 nation
```

```
dc[, c("right_rung", "why", "the_trap")]
```

```
#>                                     right_rung
#> 1 ballot
#> 2 ballot, or precinct-by-method returns
#> 3 precinct
#> 4 precinct
#> 5 precinct
#> 6 election jurisdiction
#> 7 county
#> 8 congressional district
#> 9 state
#> 10 nation
#>                                     why
#> 1 The two rankings must sit on the same ballot. No total can restore them.
#> 2   It is a joint distribution. Georgia publishes it; most states do not.
#> 3   Smallest unit with returns; demographics attach through blocks.
#> 4   County hides the variation; but check the boundary did not move.
#> 5   Fine enough to route a walk list, coarse enough to be stable.
#> 6   The jurisdiction is what administers and reports rejection.
#> 7   'County' is the unit the education measure is published on.
#> 8   The seat is the district. Precincts must be aggregated into it first.
#> 9   The electoral college is counted in states, not in voters.
```



```

#> 10      Only the top rung gives one comparable number over a long series.
#>
#> 1      Precinct returns for an RCV race give round-by-round totals, not pairs.
#> 2      Regressing precinct Dem share on precinct mail share returns 194%.
#> 3      The estimate is an inference, not a measurement. Report the bounds.
#> 4      A precinct that 'swung' may just have gained three hundred houses.
#> 5      A precinct is not a person; targeting is a bet on an average.
#> 6      'Jurisdiction' is a township in Wisconsin and a county in Texas.
#> 7      Robinson (1950): the county pattern need not hold for county residents.
#> 8      Districts do not nest into counties; someone had to disaggregate blocks.
#> 9
#> 9      The popular vote is not the quantity being decided.
#> 10     Anything below the top rung has changed shape underneath the series.

```

Step 11 — What you can now say

You can say that election data comes on a ladder, that the arithmetic of climbing is exact, and that each step destroys something specific — 30.1% of the variation between precincts, and every pair of marks that sat on one ballot.

You can say it sharper than “aggregate data is risky,” because you have a case with an answer key: 194.0% against a published 64.5%.

You cannot say which rung is correct in general. You can name the rung, say why, and check whether the answer survives being asked one rung up.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Break the regression differently. `eco_precinct.csv` has a `votes` column. Re-run Goodman’s regression *unweighted*, then weighted, then on the largest 500 precincts only, then on the smallest 500. Report the four estimates of the Democratic share of mail voters. Which one is closest to 64.5%, and does that tell you anything you could have known without the answer key?

Option B — Find the level where the answer flips. Aggregate `eco_precinct.csv` into groups of increasing size — 500 groups, 159, 50, 10 — and plot the Goodman estimate against the number of units. Where does it cross 100%? Say what that curve means for anyone choosing a unit of analysis.

Option C — Pick a question, defend a rung. Choose a research question that is *not* in `decision.csv`. Say which rung you would use, what you would lose from the rung below, and what you would risk at the rung above. Then find the number in this lab that best supports your choice.

Option D — Your own question. 17 files, 7 levels, three elections.

What this lab cannot tell you

- **How often Goodman fails this badly.** One state, one contest, one variable. We know it *can* be 129.5 points wrong and give no warning; we do not know the distribution.
- **What a cast vote record would show for an ordinary election.** Alaska’s exists because the count is ranked. Nothing here says what one would reveal about a plurality race.

- **Whether King’s method fixes it.** Not run here. `rpv` discusses it; neither lab runs it.
- **Anything about individual voters.** Every file here except the Alaska sequences is already an aggregate, and the Alaska file was aggregated to ranking sequences before it was committed.
- **Where the units are.** No boundaries, no maps. `precinct-geography` and `census-geography` are the labs about the shapes.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- The secret ballot destroys individual-level data on purpose. Should cast vote records be public anyway?
- Goodman’s regression has been used as evidence in federal voting-rights cases since the 1980s, and here it returns 194%. What should a court do with that?
- Name a claim you have read in the news this term and say which rung it was made at. Was it the right one?

Where the data came from

Every input is a CSV already committed to another lab in this course; nothing was downloaded for this one. `data/build-data.R` re-derives every number above.

- **Alaska cast vote record, 2024 US House** — `cast-vote-records/data/sequences.csv`, 340,778 ballots collapsed to 926 ranking sequences by that lab’s build.
- **Georgia precinct returns, 2020 and 2024, with the vote-method breakdown** — `ga-precinct-returns/data/`, from the Secretary of State.
- **County returns for every state, 2024** — `data-sources/data/`.
- **Presidential results by congressional district, 2024** — `redistricting/data/`.
- **National and state results, 1864–2024** — `historical-campaigns/data/`.
- **Unit counts** (177,708 precincts, 6,461 jurisdictions, 158,211,780 voters) — `eavs/data/`, the EAC’s Election Administration and Voting Survey.
- **Houston County, Georgia primary with reported race** — `rpv/data/`.